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C0-C1 joint injection: Anatomical, clinical and technical review

Christopher Mares [a,b], Carl Majdalani [b,c,*]

[a] Centre Hospitalier de l'Université de Montréal (CHUM), Montréal, Québec, Canada
[b] Institut de Physiatrie du Québec (IPQ), Montréal, Québec, Canada
[c] Hôpital Cité-de-la-Santé (CSL), Laval, Québec, Canada

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ABSTRACT

Background: Cervicogenic headaches (CGH) are proven clinical entities. The prevalence of CGH arising from the atlanto-occipital (AO) joint is unknown. The best evidence for treatment of CGH is for third occipital nerve radiofrequency neurotomy. Treatment of CGH includes intra-articular injections into upper cervical spine joints.
Objective: To perform a review of the anatomy and clinical presentation of AO joint (AOJ) pain referral as well as a technical description to safely access the AOJ.
Methods: A literature review was performed to explore the intricacies of the cranio-cervical junction (CCJ) with a focus on the relation between the AOJ and vascular anatomy. Our technical approach is described with complementary images.
Results: The AOJ lies anterior to a venous sinus and slightly superior to the horizontally oriented vertebral artery crossing the joint line.
Conclusion: The authors propose a modified superior needle trajectory that seeks to avoid these vascular structures and to access the AOJ safely.

1. Introduction

Cervicogenic headaches (CGH) are an example of referred pain. Individuals with upper neck pathologies such as whiplash injuries might complain of headaches more than cervicalgia. Neck pain can be accompanied by headaches [1]. In whiplash injuries, the C2-C3 joint is the pain generator in up to 53 % of CGH cases [2]. The best evidence for the treatment of CGH is for third occipital nerve radiofrequency neurotomy [3]. Other structures may contribute to CGH, notably the atlanto-axial (AA) lateral joint and the atlanto-occipital (AO) joints. The prevalence of the C1-2 joint as the source of pain in the CGH joint has been described [1,4], however less is known about the prevalence of the C0-1 joint. Previous work was able to identify the pain distribution of these joints thus confirming they can be pain generators [5,6].

Addressing CGH should be done with a multidisciplinary approach. Initially, a trial of physical modalities, physical therapy and analgesic and anti-inflammatory medication should be attempted prior to considering interventional approaches [7]. In the treatment continuum, intra-articular injections should be considered for diagnostic and therapeutic purposes. Efficacy has been demonstrated for mechanical and inflammatory AO and AA pain following intra-articular corticosteroids injections [8–10].

2. Objectives

In this review paper, a focus is placed on anatomy, innervation, vascularization, biomechanics, clinical presentation, diagnostic interventions and known treatments of the AOJ. We describe our current approach to AOJ injection with detailed steps and corresponding images.

3. Methods

We performed a review of the literature to cover the aforementioned topics. The objective of our research was to identify what to consider while performing AOJ injections. By understanding the anatomy, clinicians can avoid pitfalls and prevent serious complications. As our results from the review will lay out, spine interventionists should keep in mind how each anatomical component can lead to catastrophic outcomes. We share our approach to AOJ injection with images from a real case. All procedures were performed in compliance with relevant laws and institutional guidelines. Authors obtained informed and written

* Corresponding author. Institut de Physiatrie du Québec (IPQ), Montréal, Québec, Canada.
E-mail addresses: christopher.mares@umontreal.ca (C. Mares), ce.majdalani@umontreal.ca (C. Majdalani).

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consent to share images from this procedure.

4. Results

4.1. Osseous anatomy

The atlas (or C1) is unique in its morphology as it doesn't possess a body, unlike other vertebrae. The anterior and posterior arches form a ring as both are joined by the lateral masses. These osseous pillars, oriented superomedial, bear the weight of the laterally oriented occipital condyles. Therefore, C1 articulates both rostrally and caudally with the occipital condyles and the atlas, respectively. These joints allow for flexion and extension movements with the occipital condyles, while rotation is achieved via the intimate relation between C2's dens and the posterior surface of the anterior arch of C1 [11–13].

The axis (or C2) is distinguished by its vertically protruding structure, the dens. A total of 3 joints are formed between the AAJ: two lateral synovial joints and one median synovial joint complex consisting of the dens, the posterior arch of C1 and the transverse ligament. Inferiorly, it articulates with the C3 vertebra [11–13].

The occipital bone is the final part of the CCJ. Within the occipital bone, the foramen magnum serves as an exit for the medulla, which becomes the spinal cord within the neural canal. The anterior portion of the foramen magnum consists of the basion, which is a direct continuation of the clivus. Mirroring the clivus posteriorly, the opisthion is identified [11–13].

4.2. Ligamentous anatomy

The ligaments of the CCJ offer additional stability to the mobile segments of the spine [12]. Notably, the cruciform ligament, consisting of the horizontal transverse ligament and vertical longitudinal fasciculus, is the prime stabilizer of the median atlanto-axial joint. It is the largest, strongest and thickest ligament of the CCJ. The alar ligaments attach to the medial portion of the occipital condyles and depart from the anterolateral surface of the dens itself. An unpaired apical ligament attaches the dens to the clivus. The posterior atlanto-occipital membrane (AOM) originates from the posterior arch of C1 to the posterior rim of the foramen magnum. The posterior longitudinal ligament (PLL) is a direct caudal continuation of the tectorial membrane found on the posterior rim of the anterior arch of C1. Lastly, the rostral origin of the anterior longitudinal ligament is the anterior AOM connecting the clivus to the anterior arch of C1 [12,14].

4.3. Neural anatomy

The CCJ innervation is complex. Unlike its caudal counterparts, articular innervation of AOJ and AAJ is not reliant upon the medial branches of the dorsal rami. The AOJ receives direct innervation from the ventral rami (VR) of C1. Of note, it is the sinuvertebral nerve (SVN), which branches off the VR and loops back into the intervertebral canal to innervate various structures such as intraspinal ligaments, the posterior annulus fibrosis and the dura [11]. The AAJ is composed of 2 distinct joints with separate innervations. The lateral joint or C1-C2 facets are innervated by the C2 VR while their ventral portion receives neural supply from both superficial and deep cervical plexus arising from the ventral rami of C1 to C3 [15]. The medial joint of C1-C2 is innervated by the SVN of C1 to C3 [16,17].

4.4. Myology

Finally, several muscles span across the CCJ. Multiple small and deep muscles initiate the cervical movements of flexion, extension, lateral flexion and rotation. Understandably, they work in synergy with larger muscles. The suboccipital triangle is a key landmark and covers the posterior portion of the CCJ. This right-angle triangle, covered by the semispinalis capitis, is bordered superomedially by the rectus capitis major, laterally by the obliquus capitis superior, inferiorly by the obliquus capitis inferior and contains the third segment of the VA, the suboccipital venous plexus and the suboccipital nerve (or the dorsal ramus of C1) [18].

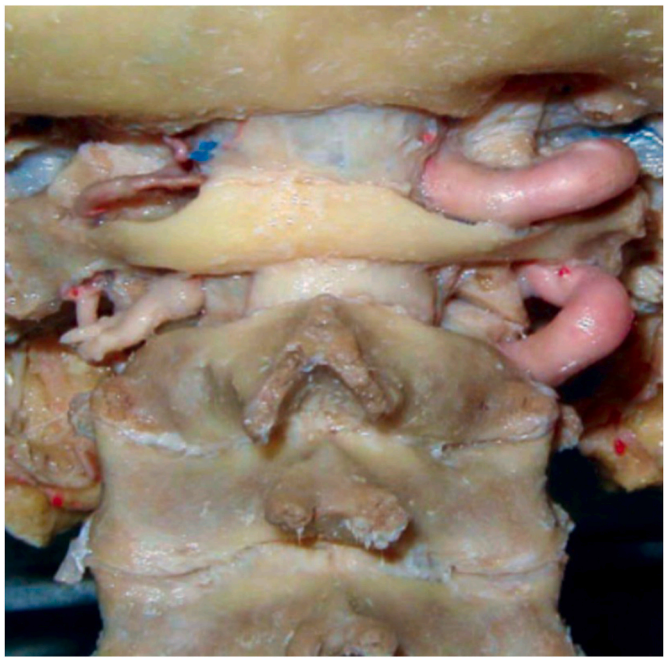


Image 1. Picture of vertebral artery course in cervical spine. The right C0-1 posterior joint line can be seen just above the vertebral artery and below the occiput (Thank you to Dr Atul Goel for letting us use this image in our paper).

4.5. Vascular considerations

The vascular supply of the CCJ can be quite variable amongst individuals. A comprehensive review and understanding of the vascular system are indispensable owing to the potential risks associated with AO procedures.

Firstly, there is a theoretical but significant risk of bleeding following either venous or arterial punctures, underscoring the importance of expert and precise needlecraft as well as anatomical knowledge. Furthermore, the risk of intra-arterial injection can lead to possible catastrophic consequences if particulate steroids are used. To minimize the risk of irreversible complications, using dexamethasone is a staple of appropriate and safe practice. Another critical consideration involves the peril of inadvertent intramural injection into arteries, which has been documented to cause arterial dissection, a complication highlighted in detailed post-mortem analyses [19]. These insights emphasize the necessity for spine interventionists to possess thorough knowledge and skill in managing and navigating vascular anatomy to mitigate risks and ensure patient safety during interventional procedures.

4.6. Vertebral artery course

The posterior cerebral circulation arises from the union of both vertebral arteries ventral to the medulla. The vertebral artery is commonly divided in four sections. Firstly, it arises, in most cases, from the subclavian artery and ascends dorsally towards the transverse foramen (TF) of C6. Secondly, it travels rostrally and penetrates each TF until it reaches C2. From there, while remaining extracranial, it projects dorsomedial behind the lateral mass of the atlas before sharply turning to pierce the dura. It completes its final leg intracranially where it merges with the contralateral vertebral artery, forming the basilar artery

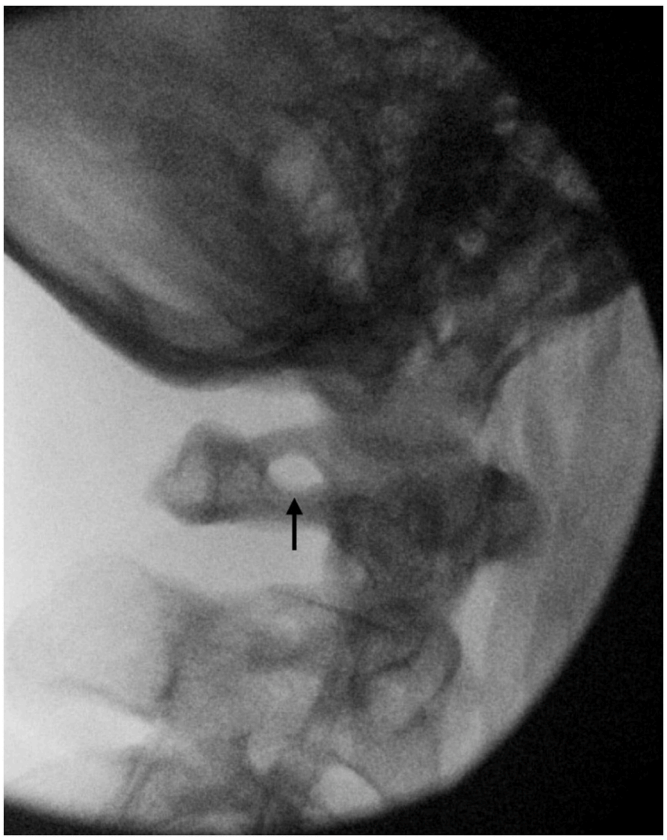


Image 2. True lateral of the cranio-cervical junction demonstrating an arcuate foramen (black arrow).

[20].

The third segment of the VA requires an in-depth analysis as the anatomy is of technical relevance to the spine interventionist. Knowledge of the location of the VA during intra-articular access of the AA and AO joints is crucial in avoiding potentially serious outcomes. One must keep in mind the three-dimensional course of the VA following its emergence from the C3 TF. The VA courses posterosuperior and is intimate with the inferior portion of the superior articular process of C3 before entering the C2 TF. It remains medial and in proximity with the C1-C2 facet as it emerges from the C2 TF [21]. In a cadaveric study by Ulm et al. (2010), the angle at which the VA exits the C2 TF is at 90° in half of the specimen. Most importantly, there is a considerable amount of variation in the trajectory in the VA's proximal loop prior to the C1 TF entry. In most cases, the VA ascends posteriorly, but it can sometimes adopt a slighter lateral course. However, it never orients medially [22].

4.7. Vertebral artery and its relation to the atlanto-occipital joint

After exiting the C1 TF, the VA courses medially and dorsally within the groove found in the posterior arch. In Cacciola et al., the VA occupies a substantial 57 % of the atlas' posterior groove width [23]. As it continues medially, the VA remains inferior to the posterior aspect of the AO joint (see Image 1) [24]. It must be known to the interventionist that the VA might overlap the joint line as it traverses it medially on the needle's path. Medial to the superior articular process of C1, the VA will pierce the AOM and make its way intracranially via the foramen magnum. Some authors have described an ossified canal formed within the AOM in which the VA travels [24]. V3 gives off multiple smaller branches such as the radiculo-medullary artery accompanying the C2 SN laterally, the muscular artery supplying the deep suboccipital musculature, and the posterior meningeal artery vascularizing the dura and the posterior portion of the tentorium [25].

Numerous anatomical variations of the VA have been reported in the literature. While planning an AOJ injection, one must carefully review three-dimensional imaging to ensure proper understanding of the VA course. Variations such as extracranial and extradural origin of the posterior inferior cerebellar artery (PICA) originating from the horizontal V3 segment has been described to travel in parallel with C1 SN and might be an additional obstacle on the needle's path [26,27]. Proatlantal arteries are rare vascular anomalies in which a persistent embryonic intersegmental artery connects the internal carotid artery to the V3 segment of the VA prior to its intradural entry [28]. On the other hand, it is possible for the VA to enter the spinal canal at C1-C2, bypassing the sinuous path seen in V3 [29].

4.8. Relationship between the arcuate foramen and the vertebral artery

Although not systematically present in everyone, the arcuate foramen (ArF) is a bony opening found on the posterior arch of the atlas. It is thought to be the result of the ossification of the posterior AOM (see Image 2) [30]. The bridge originates from the posterolateral aspect of the superior articular process and courses posteroinferiorly towards the C1 arch. The clinical and anatomical significance of the ArF is related to the passage of the VA's third segment through the foramen itself. Indeed, it has been a potential site of transient ischemia associated with vertebrobasilar insufficiency [31,32]. The ArF can be identified on simple X-Ray imagery, but ideally, a three-dimensional computed tomography will best describe the complex formed with the VA [33].

More importantly, for the spine interventionist, its presence can be identified prior to the procedure. A classification proposed by Hasan et al. (2001) suggests that variations in the ossification of the posterior AOM exist [34]. In some cases, it might hypothetically obstruct the

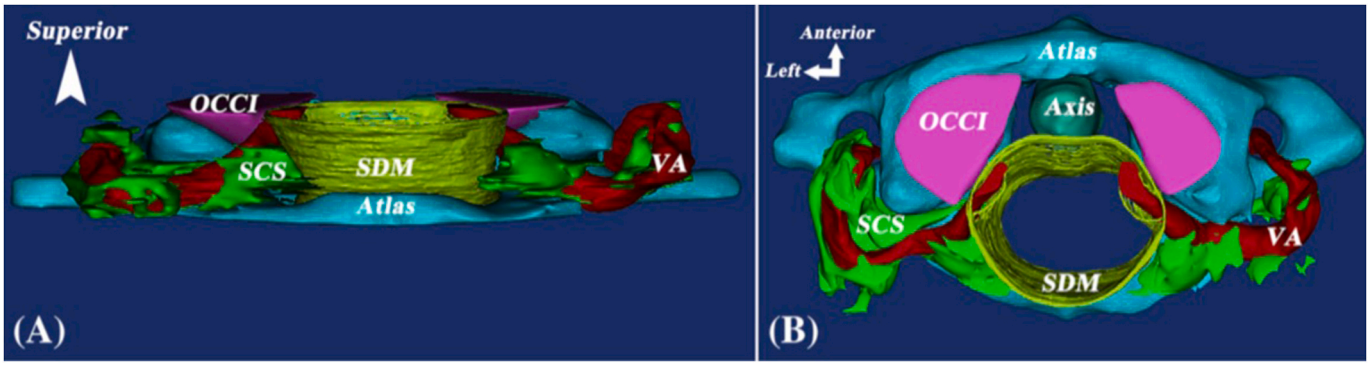


Image 3. A 3D reconstruction of the suboccipital cavernous sinus. Note relation with respect to the vertebral artery. (A) represents a dorsal view while (B) represents an axial view. SCS = suboccipital cavernous sinus; VA = vertebral artery; OCCI = occipital bone; SDM = spinal dura mater. (We would like to thank Dr Nan Zheng for allowing us to use this image in our paper).

access to the AO facet if the calcification extends superiorly. An association between high-riding VA and ArF has been suggested [35]. This concept is of importance to spine surgeons when it comes to the instrumentation of the CCJ but wouldn't influence the approach for AO joint block.

The presence of the ArF will modify the morphology of the VA with cervical rotation. Indeed, elongation and narrowing can result in a spectrum of clinical presentations ranging from transient neurological symptoms to permanent cerebrovascular sequelae involving thrombi or emboli [30].

4.9. Venous network

Less described is the vast and dense venous network surrounding the VA within the suboccipital region. Coined the suboccipital cavernous sinus (SCS), this vast venous system surrounding the V3 segment can be a significant obstacle during needle driving towards the AOJ. When describing the venous anatomy, a distinction is made with regards to the horizontal and vertical component of the VA around the AOJ.

The in-depth dissection work by Arnautovic et al. described a substantially sized, irregularly oval, venous compartment resting superiorly to the posterior arch of C1, dorsomedial to the dura, and dorsolateral to the capsule of the AOJ as well as the horizontal portion of V3 [25]. This venous bulb is surrounded by a fibrous membrane containing a periarterial autonomic neural plexus. Superficially to that, a rich and tortuous plexus is lodged between the intermediate (semispinalis capitis) and deep muscle layers of the suboccipital triangle [25]. Deeper to the venous bulb, the horizontal portion of V3 is embedded within a venous complex that contains both epidural and paravertebral veins [36]. The ascending vertical portion of V3 is surrounded by a VA proper venous plexus which extends inferiorly into vertebral veins [37]. Although beyond the scope of this paper, the dissections allowed the authors to describe a vast venous network connecting other veins (i.e. condylar) and sinuses (i.e. marginal) to the suboccipital cavernous sinus (see Image 3).

Building off previous work, novel three-dimensional (3D) visualization models allow for a greater understanding of the SCS [38]. It originates from the transverse foramen of the atlas and terminates at the dural penetration of the VA. It consists of 3 distinct portions in relation to the VA groove on the posterior arch of C1. Beginning at its lateral origin, between the transverse foramen and the VA groove on the posterior arch, the middle portion is found at the VA groove, and the medial third is past the VA groove to the dura. With the use of 3D software modeling, Zhang et al. were able to further reconstruct the internal portion of the SCS [38]. They found that the medial and lateral section of the SCS contained greater venous sinuses than the middle part. It's the latter that is situated dorsally to the AOJ. Histological studies demonstrating the presence of multiple smaller sinuses within the SCS are reinforced by fibrous bands [38].

In the neurosurgical literature, it has been stated that the SCS, the VA venous plexus and the vertebral venous plexus were the primary sources of hemorrhaging in upper CCJ surgeries [39]. The supporting collagen tissue is thin and flimsy, offering very little protection to the venous network. However, histological studies revealed a continuation of collagen fibers departing from the suboccipital triangle musculature to the cervical spinal dura. It is speculated that this interconnection modulates blood circulation during muscle contraction or relaxation [38]. The risk of important hemorrhaging is less, however never zero, when needle driving as it isn't as traumatic as suboccipital surgery. In a situation of potential hemorrhage, the low-pressure venous system of the SCS may be countered by the intracompartmental pressure of the surrounding musculature [19]. Given the size and location of these plexi, it is possible that some form of venous penetration may occur during AOJ injection.

Pre-procedure 3D imaging for planning can demonstrate anomalies in venous plexus rendering joint access riskier. An example of venous ectasia posterior to the C0-1 joint leading to the decision not to inject is portrayed in Image 4.

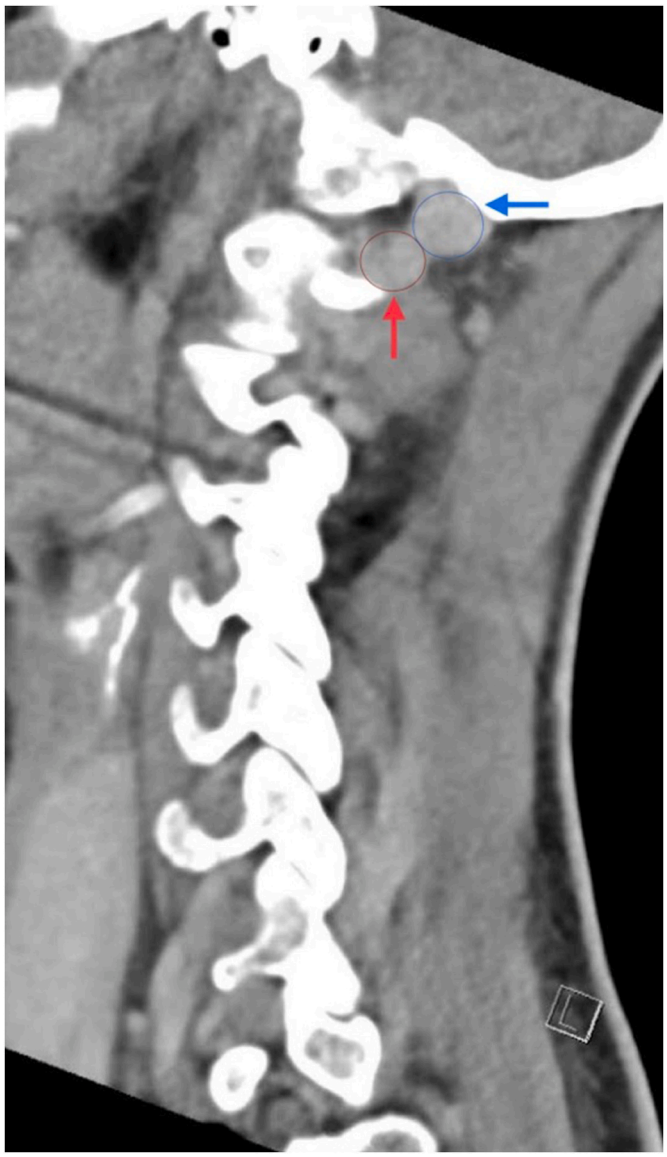


Image 4. Venous ectasia (blue arrow and circle) located superior to vertebral artery (red arrow and circle) and directly posterior to C0-1 joint. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

4.10. Relationship between cervical nerve roots and the vertebral artery

The intimate relation between the VA and neural structures must be properly understood by spine interventionists. Generally, following their exit from their respective intervertebral foramina, the nerve roots run posterior to the VA. The spinal nerve (SN) travels in a groove found on the transverse process of the inferior vertebra. The VA's 2nd segment is perpendicular to each SN as it ascends prior to becoming the third segment. Interestingly, the distance at which the VA intercepts the SN narrows rostral to C3 and tends to be measured at less than 4 mm from the intervertebral foramina [40].

At the AO segment, the VA adopts a horizontal trajectory within a groove on the superior portion of the posterior arch of C1. The 1st SN travels inferior to and, most contrary to other SN, anteriorly the VA inside that groove. As it exits the space between the occipital bone and the posterior arch of the C1, the C1 SN gives off a dorsal ramus which

supplies the suboccipital triangle musculature complex and has a cutaneous component [40]. A ventral ramus also emerges from C1 SN and continues antero-laterally to the lateral mass of C1 until it reaches the rectus capitis lateralis [18].

4.11. Biomechanics of the atlanto-occipital joint

The atlanto-occipital joint is comparable to a cup. The ovoid and slightly concave superior articular process of C1 supports the bony occipital condyles. This synovial joint is ellipsoid or condyloid in type and both joints act as one. Encompassing the joints are fibrous capsules offering reasonable stability to the articulation. The AOJ moves primarily in flexion and extension and has minimal lateral flexion and rotation. While joint excursion varies amongst individuals, it is generally accepted that a normal range of motion (ROM) pertaining to cervical flexion and extension reaches 25°. Most of the motion is achieved in extension as the odontoid process limits the flexion by contacting the anterior arch of C1. As for lateral flexion, cadaveric studies yield ROM up to 3–5°, while rotation reaches 2–7° [14,41]. Those movements are not physiological by muscle contraction and require external forces to be attained. For cervical rotation to occur at the AOJ, the condyles of the occiput must be lifted and separated from the atlas. Normally, the posterior portion ipsilateral occipital condyle will contact the posterior joint socket, thus limiting the rotation. For lateral flexion to occur at the AOJ, the contralateral occipital condyle must be lifted, which creates tension of the AOJ's capsule [42].

Spinal stability is crucial for proper CCJ health. Acquired instability can be multifactorial and of different etiology. However, active neuromuscular control and passive ligamentous tension are essential in achieving craniovertebral stability [18]. If any disruption of healthy tissue occurs (acutely or chronically), progressive instability can develop. The main component of joint stability is provided by extra-articular structure's biomechanical properties. The tissue stiffness dictates the stability. If instability develops, joint degeneration occurs and clinical findings may arise, ranging from mild pain to potential tetraplegia [14].

4.12. Physical examination of the atlanto-occipital joint

While there are no pathognomonic physical exam maneuvers, most studies have described a specific maneuver to assess for AO pain. With the examiner's thumb placed between the mastoid process and the transverse process of the atlas and the first webspace stabilizing the CCJ, the clinician uses their contralateral hand, placed on the patient's vertex, to mobilize the cranium. Ipsilateral lateral flexion and contralateral cervical rotation was used to evaluate AO mobility. Lack of range of motion or pain might indicate an AO pain generator [8].

4.13. Pathophysiology of cervicogenic headaches

In the foundational review by Bogduk and Govind (2009), the mechanism behind the pain referral stems from the convergence of afferent pain signals from upper cervical nerve roots and trigeminal nerve branches towards the trigeminocervical nucleus. The confirmation of these nociceptive signals was done by provoking, indirectly, the spinal nerves via intra-articular injections of saline in healthy volunteers. The capsular distension irritated the terminal endings of the spinal nerves in question [43].

4.14. Pain referral pattern and clinical presentation

Articular mediated pain referral patterns for AO and AA joints were initially studied on asymptomatic healthy volunteers. AO pain referral patterns were broader and of greater intensity than the AA joint. It was shown that pain remains ipsilateral. AO pain was mainly suboccipital with some spread cephalad towards the vertex and laterally towards the ear without including it. In other words, AO pain, schematically, follows a "male baldness" pattern. However, AA pain remained consistently suboccipital in all subjects [44]. Fukui et al. described slightly different findings in AO pain patterns: 100 % of the subjects experienced retro-auricular pain in the posterolateral cervical region and 30 % had occipital pain inching towards, but not reaching, the vertex. They had identical and non-discriminatory pain patterns for the AA joint as well [6]. In addition, it is hypothesized that the intimate relation between the trigeminocervical nucleus and the afferent trigeminal nerve and upper cervical spinal nerves can explain the oculo-fronto-temporal pain experienced by individuals with AO and AA pain [45]. Other than pain and headaches, patients may complain of potential neck tightness or stiffness, but also nausea, tinnitus, blurry vision, and insomnia [46].

A practice guideline concluded that history and physical exams offer no pathognomonic features in identifying AO and AA pain. Globally, AA pain remains mostly suboccipital and reproducible while AO pain has more of a heteroclite pain dispersion ranging as low as C5 to the vertex. Only an intra-articular block can properly identify the pain generator [47].

4.15. Interventional treatment and complications

The start of the treatment section must re-emphasize the inherent risk for this block related to the complex neurovascular anatomy as described above.

More recently, one randomized controlled trial (RCT) by Shin et al. compared the efficacy of intra-articular AO injection versus intra-articular pulsed radiofrequency. 23 patients with suboccipital pain underwent AO joint diagnostic blocks following enrollment. Patients either were placed in an IA AO joint injection with corticosteroids group or an IA AO joint pulsed radiofrequency group. While there was no superior treatment, both groups demonstrated a significant reduction in pain on the Numeric Rating Scale (NRS) at follow-up until 6 months [48].

Lee and al. sought to assess the efficacy of IA AO injection in a prospective observational study. 70 % of the treated patients had complained of CGH. Following a successful diagnostic block (more than 50 % relief after 30 min) with 1 cc of 2 % Lidocaine, patients were enrolled to undergo 2 AOJ injections at 1 week interval with 0.75 cc of 2 % Lidocaine and 0.25 cc of Triamcinolone 10 mg following IA confirmation via contrast product. Results showed a statistically significant drop in pain on VAS and improvement on Neck Disability Index (NDI) score. While follow-up was only done up to 2 months, they demonstrated the efficacy in short-term relief [8].

Two studies by Busch et al. and Dreyfuss et al. demonstrated relief in AO joint injection in a total of 28 patients suffering from CGH [5,49]. These studies were not RCTs. Centeno et al. mentioned having successfully treated 10 patients as well [50].

Notwithstanding the adverse and potentially fatal events with any spine intervention relating to particulate steroid use intravascularly, AOJ injections have their own risks associated with needle driving. As mentioned earlier in the vascular section, the VA's V3 segment lies posterior and, generally, inferior to the AO joint line. However, any anatomical variant can misplace the VA on the needle's path. The suboccipital venous system can potentially completely obstruct AOJ entrance. The neurovascular consideration requires proper preparation and case-by-case evaluation of the patient's CCJ with additional imagery, such as an MRI. Although no complication with AOJ injection has been described in the literature, to our knowledge, the inherent risk should be considered significant. Consequently, this block should not be done unless significant prior training, experience and preparation as described below.



Image 5. Right vertebral artery angiogram course around the right C0-1 joint (black arrow). (Thank you to Dr Paul Dreyfuss for personally sharing the original image and allowing us to use it in our paper.).

5. Discussion

5.1. Atlanto-occipital joint approach

With the above anatomical literature review, we wish to share our technique for intra-articular access of the C0-1 joint. This may be for delivery of local anesthetic for diagnostic blockade or medication for pain relief (ex: corticosteroid, platelet rich plasma, etc).

We agree with published guidelines that 3D imaging to plan for this technique is warranted. A CT-scan with venous contrast is our modality of choice due to high spatial resolution and the combination of dedicated MRI of the upper cervical region. These images should be reviewed by a skilled and trained interventionist to determine angle and approach for safe access to the posterior part of the C0-1 joint. We understand that access might be limited for interventionist in a private practice. However, considering the potential harms associated with arterial or venous puncture, ensuring all precautions are taken prior to the injection is primordial in mitigating serious and potentially fatal complications.

As for the cessation of anticoagulants or antiplatelet agents prior to AOJ injection, shared decision making as well as weighing risks of a thrombotic versus hemorrhagic event in patients must be carefully taken into consideration. Given the extensive venous plexus described, the risk of bleeding could theoretically be higher than for the AAJ.

Our approach is adapted from Furman and Centeno with some modifications and detailed steps [50,51]. The general goal is to access the posterior supero-lateral part of the joint and adapt this to patient-specific neurovascular anatomy. In Image 5, Dreyfuss shows the relation of the VA to the joint.

Steps:

1. Preprocedural planning
   a. Review 3D images

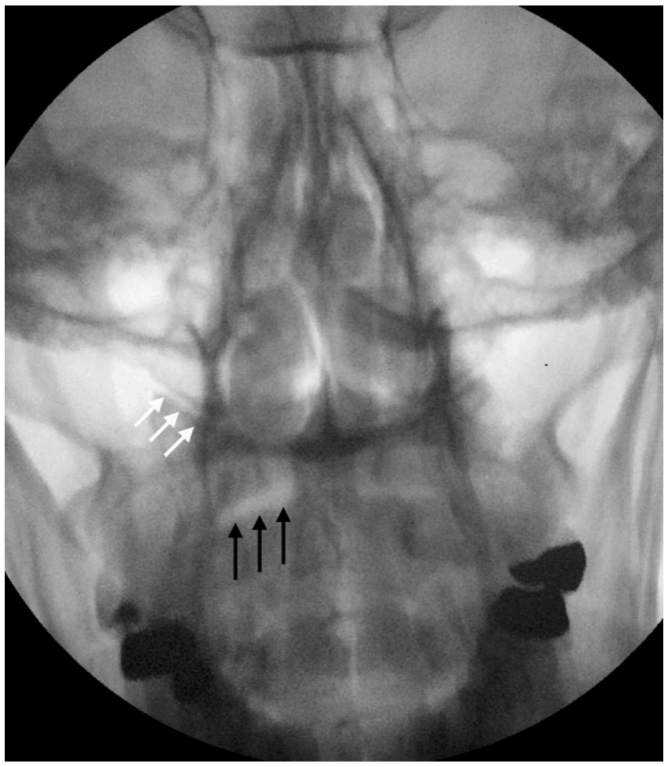


Image 6. AP view to recognize the left C1-2 (black arrow) and C0-1 joints (white arrow).

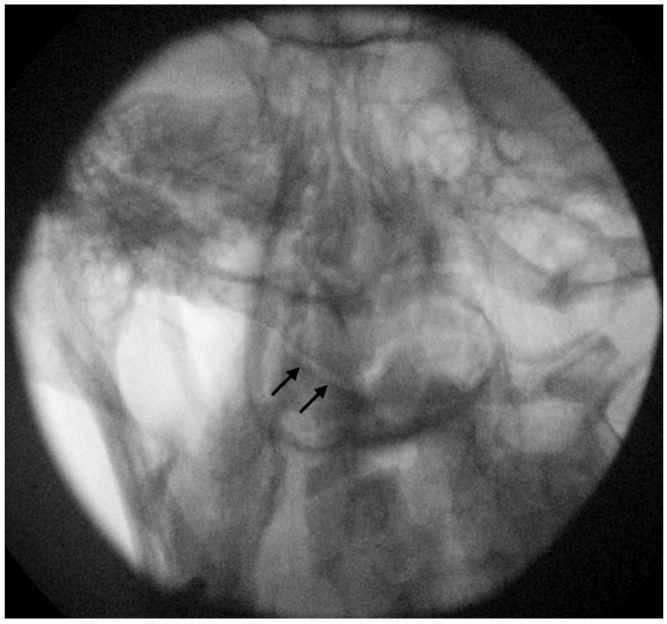


Image 7. Ipsilateral oblique to left by approximately 15° to view left C0-1 joint (black arrow).

   b. Patient is placed in prone with slight neck flexion and no rotation
      i. We choose no rotation as opposed to the Centeno technique to be able to obtain a true lateral. Also, 3D imaging is obtained in the neutral position.
   c. Obtain AP view with fluoroscope (Image 6)
      i. The dens is easily recognized as well as the C1-2 lateral joint

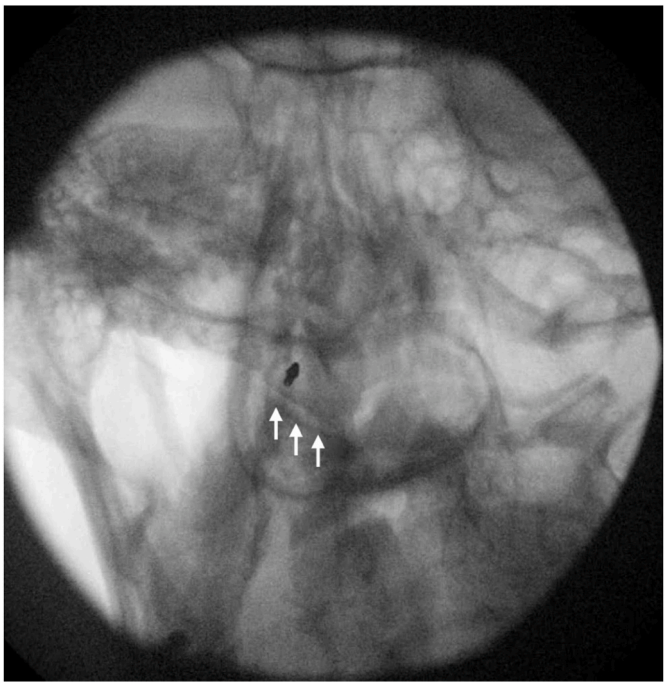


Image 8. In AP view, the target point in supero-lateral aspect of condyle (as shown by the pen tip) which is above the joint line (the joint line corresponds to the bottom of the cup of the C0-1 joint (white arrow)).

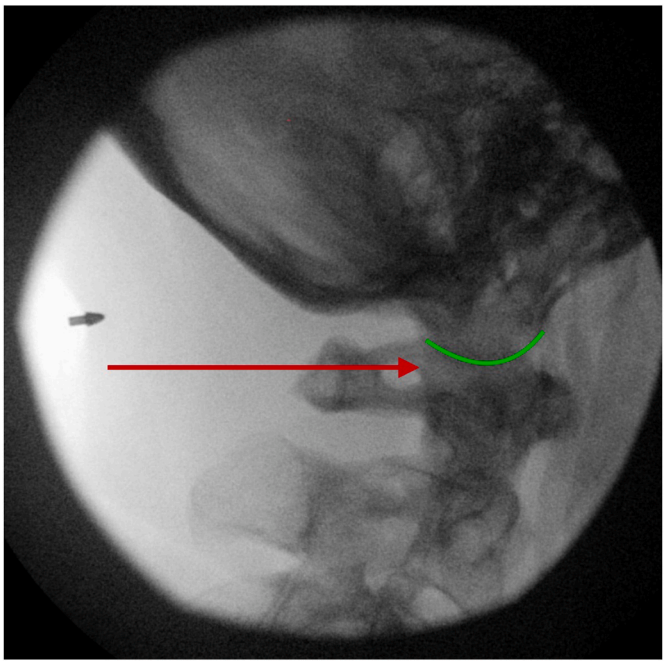


Image 9b. The red line represents the trajectory one would adopt if they were to target the C0-C1 joint line as viewed in AP. Notice how the needle driving could pass directly through the V3 segment of the vertebral artery. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

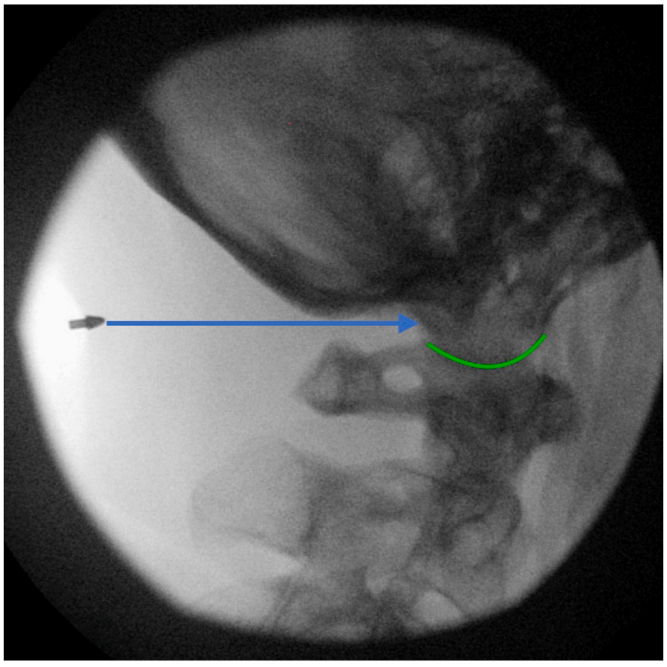


Image 9a. In this true lateral view, the pen tip is oriented above the C0-C1 joint line (green) as to bypass the vertebral artery. The blue line represents an ideal and safe trajectory. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)



Image 10. Target point with 25G spinal needle, high above joint line.

d. Oblique ipsilateral approximately 15–20° to visualize the C0-1 joint (Image 7). It is generally sloped inferomedial to superolateral.
e. Cephalo-caudal tilt is used to move the occipital brim out of the way
f. An ovaloid image should be obtained superior to the joint line - this will be the target and corresponds to the occipital condyle.
g. Mark target supero-lateral to the joint (Image 7) and use a pen to mimic hub-view. The joint line seen on fluoroscopy represents the bottom of the cup; the target point is higher than this (Image 8)
h. Obtain a true lateral while maintaining pen position and ensure pen trajectory aligns with desired target and is high above the posterior arch of C1 and just below occipital (Image for theoretical trajectory) (Image 9)

2. Procedure
   a. Disinfect vigorously especially in the setting of significant hair



Image 11. Hub view in AP, high above the joint line.



Image 13. Advancement of needle in hub view.

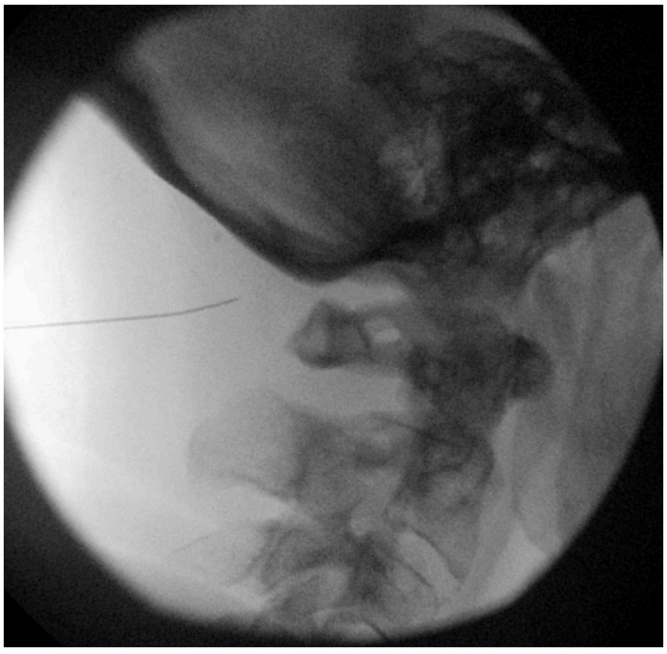


Image 12. Multiplanar view to verify trajectory, aiming above the joint line.

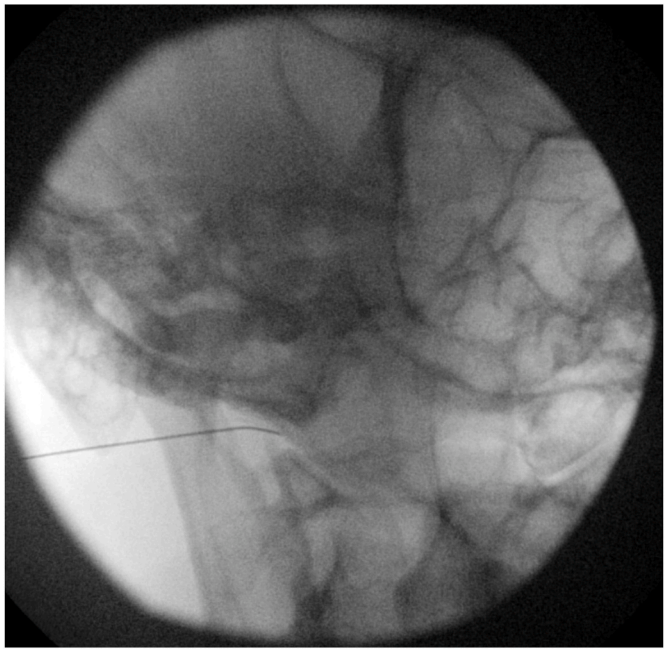


Image 14. Contralateral oblique view at joint arrival.

b. Material
   i. 25G spinal needle with slightly accentuated bevel for needle driving
   ii. Contrast agent with extension tubing
   iii. 5 mg dexamethasone and 0.5 cc 2 % lidocaine for diagnostic part of the block
c. Trajectory view
   i. Patient prone as above
   ii. Ipsilateral oblique and cephalo-caudal tilt as above to clear the joint
   iii. Choose target above the actual joint line as the joint seen is the bottom of the cup of the joint (Image 10)
   iv. Obtain hub view (Image 11)
d. Multiplanar view
   i. Once initial tissue purchase obtained with hub view, obtain a lateral view to confirm trajectory is high above C1 arch (Image 12)
e. Return to trajectory view
   i. Return to AP for further advancement in trajectory view (Image 13)
f. Multiplanar view
   i. Return to lateral to confirm once again the trajectory is staying high and not straying toward the VA on the posterior part of C1 arch
      1. With experience this step may be omitted
g. Trajectory view
   i. Final advancement done in AP in trajectory view until osseous contact.



Image 15. Arthrogram in contralateral oblique view.



Image 16. Arthrogram in lateral view, highlighting the cup shaped joint.

h. Multiplanar view
   i. Use of the contra-lateral oblique view can be helpful to use bevel and obtain intra-articular access (Image 14)
   ii. Contrast under live fluoroscopy is done in this view and confirmed in lateral and AP view
      1. Lateral arthrogram demonstrates the cup-shaped or bean shaped joint as previously described
      2. Arthrograms (Images 15–17)

6. Conclusion

CGH are a source of pain, morbidity and loss of quality of life for patients. It is well known that the structures involved are the C2-3 facet joint followed by the C1-2 and C3-4 facet joint and then the C0-1 joint. Although less common, the C0-1 joint can be a source of pain, and it can be an interventional target. In depth understanding of the anatomy in this region is crucial before any clinician should consider this procedure as well as experience and well-developed needle craft skills. Proper 3D pre-procedure imaging should also be obtained and analyzed before this procedure. Although the risks in this region are more significant, the potential therapeutic and diagnostic value of this procedure for patients can be significant and therefore we described our approach and technique.

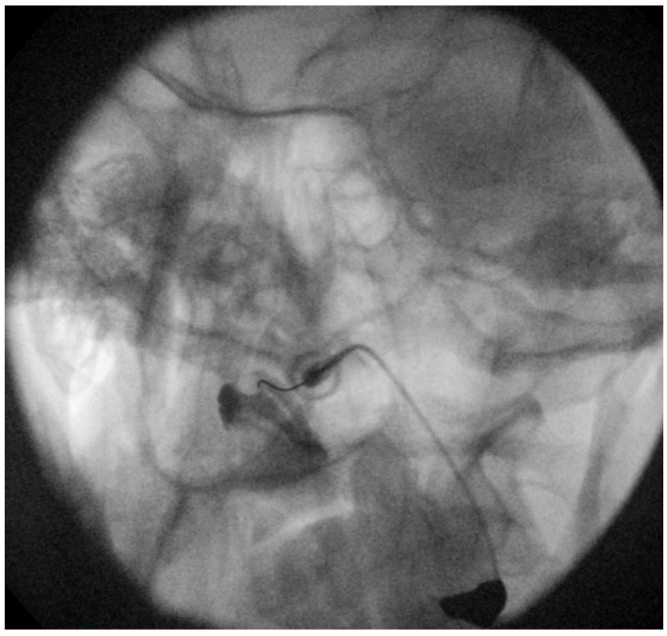


Image 17. AP view arthrogram.

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Declaration of competing interest

Carl Majdalani: None to declare. Christopher Mares: None to declare.

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